

Article

A Two-Stage Optimization Model for Airport Stand Allocation and Ground Support Vehicle Scheduling

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Abstract: To address the issues of inefficient resource allocation and severe ground congestion at hub airports during aircraft turnaround operations, a two-stage optimization model is constructed to coordinate the scheduling of stands and ground support vehicles. The model focuses on analyzing the scheduling rules, operational patterns, and collaborative mechanisms between stand allocation and ground support vehicles, taking into account the coupling relationship between airport operational and support resources. The pre-allocation of stands is conducted under the constraints of limited support resources, and the results are used as inputs for ground support vehicle scheduling. This combined optimization of stands and vehicle resources enhances the overall resource efficiency. The NSGA-II algorithm, combining local search strategies (LS-NSGA-II), is used to solve the model. Computational experiments conducted at Shenzhen Airport show some improvements: For the stand allocation model, the model incorporates ground support service constraints for tow tractors and driving distances for ferry buses, thereby avoiding potential service conflicts and resource wastage. Secondly, for the scheduling of vehicles, by analyzing the operational patterns and service characteristics of different vehicles, the model improved vehicle utilization efficiency by 37.5%, reduced travel distance by 20.4%, and decreased waiting times by 57.6%, compared to the first-come-first-served strategy currently employed at airports.



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Keywords: airport operations; stand allocation; ground support vehicles scheduling; collaborative mechanism

1. Introduction

With the rapid development of the civil aviation industry, the increasing flight density at major hub airports has led to a shortage of available resources. In 2023, airports in the Pearl River Delta region handled a total of 936,300 flights. Among them, Shenzhen Bao'an Airport handled 393,100 flights, with an on-time departure rate of 88.67%—a year-on-year decrease of 6.72 percentage points. As a critical area for aircraft parking and ground support services, the stand has become a significant factor restricting the growth of flight volume. Improper allocation may result in decreased passenger satisfaction, and low airport resource utilization efficiency. Additionally, inefficient scheduling of ground support vehicles can further degrade flight service quality. The rapid growth in demand for civil aviation has placed new demands on the efficiency of airport support services and resource supply capacities. The civil aviation development plan highlights that the issues of insufficient capacity and low efficiency remain significant. Additionally, the imbalance and inadequacy in civil aviation development are still prominent. These issues are mainly reflected in the shortage of key resources and the dual bottlenecks of capacity and efficiency in infrastructure support capabilities. There are two primary approaches to addressing this challenge: one fundamental solution is to expand airport infrastructure

to increase capacity, while the other involves improving resource utilization efficiency to enhance airport throughput [1], which is the focus of this study. Airport resources include operational resources and support resources; this study focuses on stands and various ground support vehicles.

Stands can be categorized into contact stands and remote stands. At busy airports, contact stands, which connect to terminals via jet bridges and are equipped with comprehensive turnaround service facilities, are in high demand. The core goal in the stand allocation problem is to assign available terminal or apron positions to each flight, minimizing passenger inconvenience and maximizing airport operational efficiency [2]. How to complete multiple support services for flights within a limited timeframe is one of the more complex challenges airports face. Currently, most airports adopt a first-come-first-served (FCFS) manual scheduling approach for vehicle dispatching, which means the ground support vehicles that complete their tasks first are assigned to the corresponding flights based on the flight arrival time. However, due to the limited availability of vehicles and the inability of manual scheduling to account for overall dynamics, the efficiency of vehicle scheduling is relatively low. Figure 1 shows the stand occupancy distribution at Shenzhen Airport over a single day. It can be observed that, due to the presence of overnight flights, the number of occupied stands is highest between 11 PM and 6 AM the following day. C-type and E-type aircraft dominate in number, and during certain periods, the high volume of C-type flights necessitates the use of combined stands, thereby reducing the availability of E-type stands. From 10 AM to 9 PM, the overall stand usage exhibits a dynamic balance, characterized by a high turnover rate and frequent stand usage during this period.

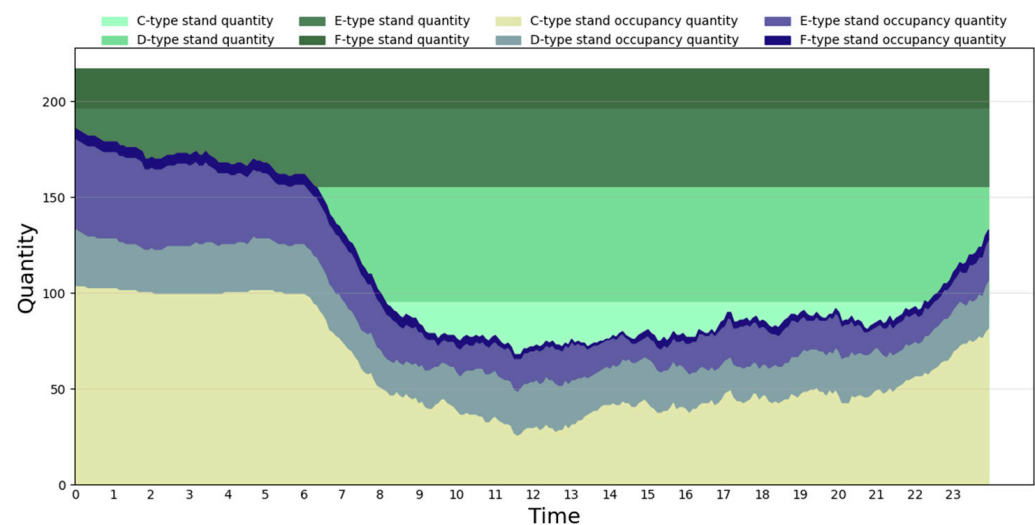


Figure 1. Shenzhen airport stand occupancy situation.

Stand allocation problem and ground support vehicle scheduling problem involves a complex interaction of multiple resources and stakeholders. The stand allocation results directly affect the scheduling of ground support vehicles, and there are coupling relationships between different vehicle types. The use of some vehicles like tow tractors may impose indirect constraints on stand allocation. Therefore, optimizing the stand allocation and ground support vehicle scheduling for turnaround flights is of significant practical importance. The main contributions and findings of this study are as follows:

(1) The study focuses on the collaborative scheduling of operational and support resources during turnaround operations. Under the constraints of specific areas and limited resources, an optimization scheme for stand allocation and vehicle scheduling is proposed, with results provided within an acceptable time frame for both stand assignments and vehicle utilization.

(2) A comprehensive analysis of the coupling mechanism between operational and support resources is conducted, and two optimization models are constructed to address the scheduling of stand resources and support vehicle resources, respectively. A two-stage optimization framework is proposed, which integrates ground operational resources with support resources. The stand allocation results are used as inputs for the ground support vehicle scheduling model, enabling a global optimization of airport ground operations.

(3) A highly efficient NSGA-II algorithm, enhanced with local search strategies, is designed to solve the models. Experiments based on operational data from Shenzhen Airport demonstrate that the multi-apron, multi-vehicle type airport resource scheduling optimization model significantly improves operational efficiency.

The remainder of this paper is organized as follows. Section 2 reviews the current research on the stand allocation problem and ground support vehicle scheduling problem, highlighting the limitations of existing studies. Section 3 constructs a stand allocation model that considers the constraints of support resources, and Section 4 builds a ground support vehicle optimization scheduling model based on the stand allocation results, considering the service patterns and coupling relationships of the vehicles. Section 5 explains the computational process and key steps of the LS-NSGA-II algorithm. Section 6 presents an experimental case study using operational data from Shenzhen Airport, offering an in-depth analysis of the optimization results. Section 7 concludes the paper with a summary of findings and future research directions.

2. Literature Review

Currently, extensive research has been conducted on the stand allocation problem and ground support vehicle scheduling problem. The following section reviews the latest research on stand allocation and ground support vehicle scheduling.

The stand allocation problem (SAP) involves assigning aircraft activities (arrivals, departures, and intermediate parking) to aircraft stands (parking positions) with the objective of maximizing the number of passengers/aircraft on contact stands and minimizing the number of towing movements, while meeting a range of operational and commercial requirements [3]. This problem is closely related to the gate allocation problem (GAP). This literature review focuses on studies related to the GAP. Steuart [4] was one of the first scholars to introduce this problem, and since then, many researchers have conducted in-depth studies on GAP variants. The stand allocation problem becomes a complex NP-hard problem due to complex airport layouts, high numbers of flights, high flight frequency, and significant turnover rates, which cause many complex factors in airport gate assignment. In most cases, this global optimization problem is nonlinear and discrete. With the development of airports, the complexity of gate allocation grows exponentially and cannot be solved by traditional mathematical methods. Therefore, there is an urgent need for an intelligent allocation method. The main methods currently in use are expert systems [5], exact algorithms [6,7] and heuristic algorithms [8–12].

Stand allocation affects three main stakeholders: airport operators, airlines, and passengers [13].

Passenger demands primarily include efficient and smooth check-in processes, short walking distances within the airport, and convenient access to airport facilities such as restaurants, shops, and entertainment options. Maharjam and Matis [14] defined a cost coefficient that considers both the walking distances between connecting flights and the transfer times between flights. Benlic et al. [15] aimed to minimize the number of passengers who might miss their connecting flights by calculating the product of the number of transfer passengers and the time required to travel from the arrival parking to the departure gate.

For airports and airlines, congestion primarily occurs in the apron area, taxiways, and runway heads, making the optimization of these bottleneck resources a key focus in the study of stand allocation problems. When flights cannot be assigned to contact stands, passengers must use ferry vehicles to travel back and forth between the stand and the terminal, which increases waiting times and decreases passenger satisfaction, particularly

during adverse weather conditions, while also raising the operational costs of airport ground support resources. A common optimization objective is to minimize the number of aircraft assigned to remote stands [16]. Improving the utilization efficiency of contact stands can also reduce ground operation costs; if the parking positions for arriving flights differs from those for departing flights, it necessitates the use of towing vehicles to move aircraft, thereby incurring additional high operational costs. Yu et al. [5] proposed that the arrival and allocation of stands are distinct activities and are aimed to minimize the secondary objective function of task-towing costs.

The stand allocation problem is currently managed by using pre-allocated stands based on the flight schedule for the following day. However, issues such as delays, flight cancelations, and resource shortages during actual operations can interfere with these pre-allocated results, making robustness very important for stand allocation outcomes. A common method to enhance robustness is to set buffer times between adjacent flights assigned to the same stand. The time interval between two consecutive uses of a gate is referred to as the gate's idle time or slack time [17]. Kumar and Bierlaire [18] penalized idle times that exceed a predetermined threshold, which is derived from historical flight delay statistics. Dijk B et al. [19] employ the concept of recoverable robustness to obtain a recoverable robust solution for the gate assignment problem. It discusses and tests four objective functions to evaluate the efficiency of the gate assignment plan.

The ground handling of flights involves coordinated collaboration among airports, airlines, and air traffic control, with flight arrivals and departures being dynamic and subject to time constraints. The coordination among various ground support vehicles follows a specific sequence and is subject to resource limitations. The primary task of ground handling is to assign ground support vehicles and operational time slots to aircraft within the constraints of vehicle resource limitations and standardized ground handling procedures. Fink M et al. [20] considered the synchronous unloading of baggage or fueling services by vehicles and workers, developing a branch-and-price heuristic method that employs traditional variable branching and innovative variable-fixing strategies. Two mathematical multi-commodity flow formulations based on spatiotemporal networks were proposed to effectively model five types of synchronization, including mobility and load synchronization. Guo W et al. [21] introduced a collaborative mechanism in genetic algorithms for the scheduling problem of airport baggage vehicles. Zhang Feng et al. [22] formulated the scheduling problem of airport water vehicles as a vehicle routing problem (VRP), integrating the hard time windows of unmanned water vehicles servicing flights with trapezoidal fuzzy membership functions to construct flight service level functions, considering the timing rules for unmanned water vehicles servicing airport flights and further optimizing sub-paths that have not reached service capacity limits based on the C-W savings algorithm. In recent years, bionic intelligent algorithms have been extensively studied in the theoretical research of vehicle scheduling problems. These include Tabu Search algorithms [23,24], Particle Swarm Optimization [25], Simulated Annealing [26], Ant Colony Optimization [27] and Fireworks Algorithms that mimic the natural variation patterns of fireworks [28]. These algorithms have been continuously refined and improved in terms of enhancing computational accuracy, reducing computational time, and minimizing computational space. Additionally, various algorithms have been integrated to create hybrid approaches [29], leading to many solutions tailored for increasingly complex real-world logistics problems.

The aforementioned studies mainly focus on the pre-allocation phase of tasks, while research on adjusting vehicle scheduling strategies in response to changes in flight information has gradually become a hotspot in recent years. For instance, changes may occur in the arrival and departure times of service flights [30] or in the parking stands of flights [31]. Consequently, on the basis of ensuring vehicle scheduling, it is crucial to adjust vehicle routes promptly according to updated information. Such dynamic vehicle routing problems can draw on relevant studies of dynamic vehicle routing problem (DVRP) models [32,33].

The above-mentioned literature has carried out some preliminary studies on the stand allocation problem and the optimization of ground support vehicle scheduling, and some research achievements have been made. However, it should be noted that the current research has the following shortcomings:

1. For the stand allocation problem, the utilization of remote stands is relatively simplistic, with the focus solely on minimizing the number of flights assigned to remote stands. This approach does not fully reflect the real operational conditions of airports, and it lacks consideration of the differences between aircraft types and flight categories.
2. Research on ground support vehicles typically focuses on a single type of service vehicle or two interrelated types of vehicles. Few scholars have comprehensively considered multiple types of support vehicles within airports to develop integrated scheduling models.
3. Existing research tends to separate stand allocation and ground support vehicle scheduling, overlooking the fact that flight turnaround is a unified, multi-stakeholder collaborative process. As the location where ground support services are provided to aircraft, different stand allocation schemes directly affect the optimization of support vehicle scheduling. Additionally, given the limited availability of ground support resources, stand allocation should take into account the service capacity within a given timeframe to avoid flight delays caused by waiting for vehicles, which could disrupt the overall turnaround process.

From the literature review, we conclude that there is a lack of coordinated scheduling mechanisms between airport ground operations and support resources. Moreover, current studies have not fully addressed the collaborative mechanism between different types of stand allocation and ground support vehicles scheduling. Therefore, in this paper, based on the shortcomings of existing studies mentioned above, we propose improvements from the perspective of integrated airport scheduling and explore solutions to optimize resource allocation.

3. Problem Description

A transit flight can be viewed as a pair of arriving and departing flights. When arriving, flights land sequentially on runways as directed by air traffic control, then they taxi off the runway to the assigned parking position. After reaching the parking position, the ground support service starts. Various ground support vehicles arrive at the parking area according to regulations. The turnaround process involves several stages. For contact stands, jet bridges are connected, and for remote stands, passenger stairs are used. Once passengers disembark, fueling, loading and unloading of cargo, water replenishment, wastewater disposal, and cabin cleaning services are performed. After all services are completed, passengers board, and after the boarding information is confirmed, the aircraft doors are closed. Once the turnaround services are fully completed, the flight awaits instructions from air traffic control. Depending on the stand type, the aircraft may either be towed out or taxi out on its own to the runway holding point, to prepare for departure.

The tow tractor operates within an apron area, and it is typically responsible for 3–4 stands within that area. Since towing is the final stage of ground handling, if the departure times of flights assigned to the same tow tractor are very close, there may be a delay in towing services, leading to flight delays. Therefore, when allocating stands, the availability of tow tractors must be considered to avoid scheduling flights with similar departure times for the same tow tractor, which could cause service conflicts. Due to the large size and longer service time of passenger stairs, they are generally fixed to a specific apron area and rarely serve across different apron areas. Other ground support vehicles are not strictly limited in their operating zones. Due to the constraints of airport resources, vehicle allocation plans must be based on the stand allocation results. The distance between tasks affects the efficiency of vehicle services, excessive distance between adjacent tasks may result in flight delays due to waiting.

Figure 2 illustrates the three working modes of ground support vehicles. The first is the continuous working mode, where ground support vehicles depart from the depot, reach the assigned parking position to service the aircraft, and then proceed to the next parking position after completing the task. Vehicles operating in a continuous working mode include towing tractors, passenger stairs, and baggage belt loaders. The second is the resource-limited working mode, where vehicles, after servicing some flights, must go to a resource supply place to replenish resources before continuing service, as they are unable to meet the demands of the next task. Vehicles in a resource-limited working mode include water trucks and catering trucks. The third is the round-trip working mode, where vehicles need to shuttle between the service place (such as terminals or baggage handling centers) and stands while providing services to the aircraft. Vehicles in a round-trip working mode include ferry vehicles and baggage tractors [34].

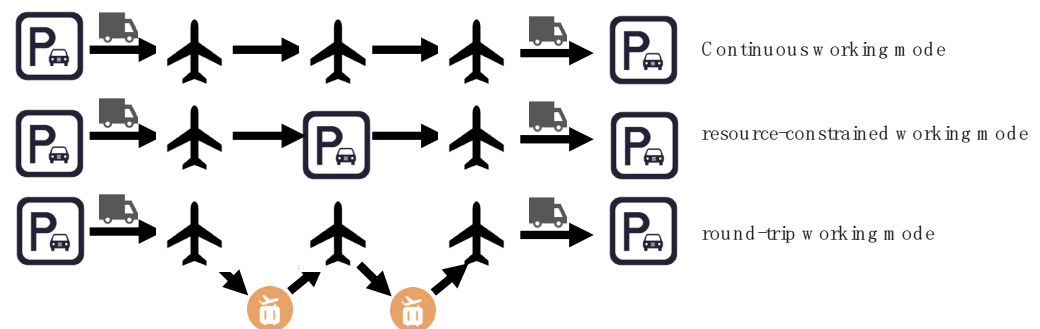


Figure 2. Diagram of ground support vehicle working modes.

A physical flight may need multiple ground support vehicles; in order to realize the one-to-one association between flight and vehicles, virtual flights are introduced.

4. Modeling

4.1. Two-Stage Optimization Model for Stand Allocation and Ground Support Vehicle Scheduling

We propose a collaborative scheduling mechanism for stand allocation and ground support vehicle resources, as illustrated in Figure 3. In the process of solving the stand allocation problem, tractor service constraints are incorporated to prevent the departure times of flights stationed in the same tractor service area from being too concentrated. Additionally, one of the goals is to minimize the driving time of ferry vehicles. The allocation results yield the assigned parking stands for each flight and the service time windows. The ground support vehicle system does not operate independently, and the gate assignment plan may lead to vehicle scheduling plans failing to meet service demands. Therefore, we adopted the two-stage model to study the integrated scheduling of ground support vehicle resources and gate resources. This approach analyzes the coupling factors and strategies of integrated scheduling.

The optimization scheduling of ground support vehicles can be understood as maximizing the task efficiency of support vehicles based on the optimal allocation of stands. This is achieved by utilizing the flight stands and service time windows generated from the allocation results. The vehicle routing schedule is then obtained according to the sequence of vehicle services and their collaborative relationships.

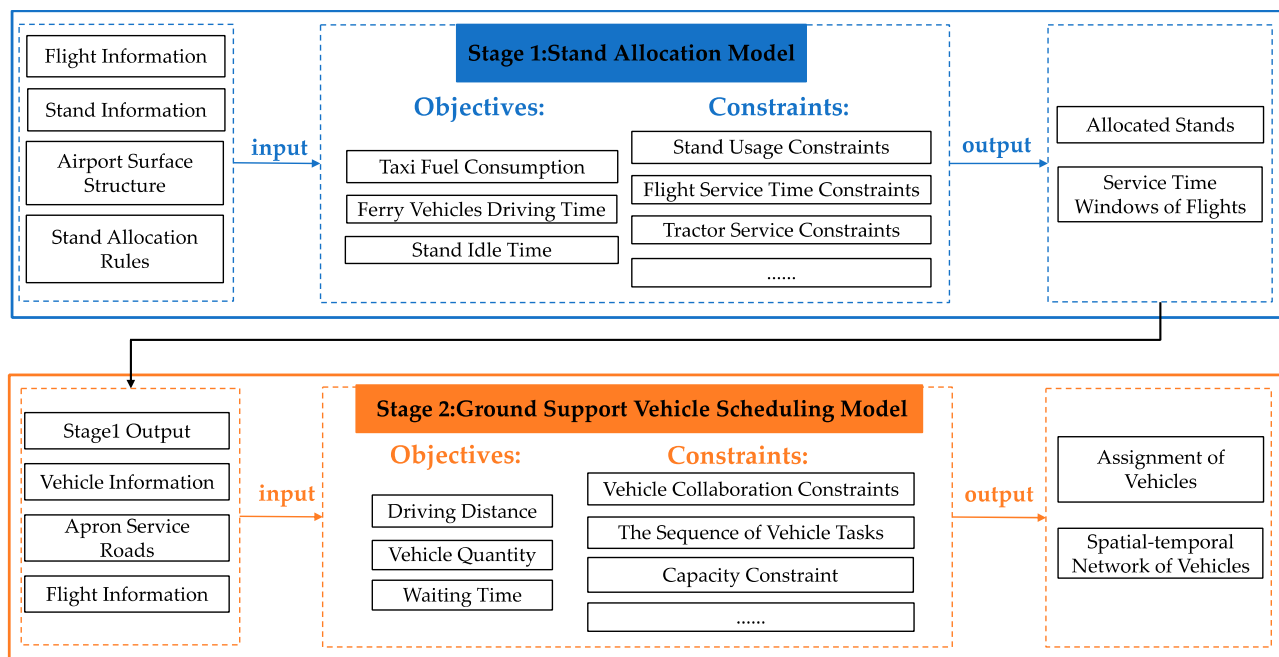


Figure 3. Two-Stage optimization model for stand allocation and ground support vehicle scheduling.

4.2. The Stand Allocation Problem Considering Turnaround Resource Constraints

The sets, parameters and variables of the stand allocation model are shown in Table 1.

Table 1. Sets, parameters, and variables of the stand allocation model.

	Notation	Meaning
sets	R	the set of runways: 0 represents Runway 15, and 1 represents Runway 16.
	F	the set of flights
	G	the set of parking positions
parameters	t_{rmg}^a	the taxiing time from runway r_m to parking position g of departure flight
	t_{grn}^d	the taxiing time from runway r_n to parking position g of arrival flight
	S_i^F	the aircraft size of flight i , $S_i^F \in \{1, 2, 3\}$, corresponding to aircraft types C, D, and E.
	v_g	the time taken for the ferry vehicles to travel back and forth between the parking position g and the terminal.
	T^i	The turnaround time of flight i
	Sa_i	The arrival time of flight i
	δ^g	the minimum safety interval for the stand g
	θ^t	the service interval of the towing tractor
	Sd_i	The departure time of flight i
variables	x_{ig}	binary decision variable, equal to 1 if flight i is assigned to parking position g , otherwise equal to 0
	y_{ijg}	binary decision variable, equal to 1 if flight j is assigned to parking position g immediately after flight i , otherwise equal to 0
	z_{ijt}	binary decision variable, equal to 1 if both flight i and flight j are assigned to apron t , otherwise equal to 0
	I_{ij}^2	the time interval between the departure time of flight i from the parking position and the arrival time of flight j at the parking position

4.2.1. Objectives of the First-Stage Model

1. Minimize the Taxi Fuel Consumption for Arriving and Departing Flights.

The objective function is shown in Equation (1). When flights are assigned to different stands, the taxi-in and taxi-out time required vary. Excessive taxi time leads to increased fuel consumption, decreased passenger satisfaction, significantly higher operating costs for airlines, and intensified environmental pollution caused by fuel emissions. Therefore, optimizing taxi fuel consumption is crucial to achieving a balance between corporate profitability, operational efficiency, and social responsibility. Aircraft taxi fuel consumption is primarily influenced by the aircraft type and taxi time. Xiong Jie [35] analyzed the average taxi fuel consumption of different aircraft types in his study. This paper focuses on C, D, and E class aircraft, and the relevant data are summarized in Table 2.

$$\min Z_1 = \sum_{i \in F} \sum_{g \in G} x_{ig} (t_{rmg}^a + t_{grn}^d) \quad (1)$$

Table 2. Average taxi fuel consumption for different aircraft types.

Aircraft Types	C	D	E
Average taxi fuel consumption (kg/min)	11.5	18	35

2. Minimizing the Ferry Vehicles Driving Time for Remote Stands.

The objective function is shown in Equation (2). During peak hours, due to the limited availability of contact stands, there are more flights than these stands can accommodate. To enhance the robustness of stand allocation and reduce aircraft waiting time on the apron, some flights are allocated to remote stands. Previous studies often used the gate utilization rate as an optimization objective. In practice, the number of ferry vehicles required for different aircraft types at remote stands varies, and the driving distance from the remote stand to the terminal stand also differs. Therefore, efficiently utilizing remote stands is equally important in stand allocation problem. Optimizing for the shortest ferry vehicles driving time to and from remote stands can indirectly reduce the number of flights allocated to remote stands, particularly larger aircraft, this also aligns with the rule of preferentially assigning large aircraft to contact stands.

$$\min Z_2 = \sum_{i \in F} \sum_{g \in G} x_{ig} S_i^F v_g \quad (2)$$

3. Minimizing the Sum of the Squares of the Idle Times.

The objective function (hereinafter referred to as idle time) is shown in Equation (3). Minimizing the sum of the squares of the idle times can, on the one hand, effectively improve the utilization rate of all stands, avoiding the overuse of some stands while others remain relatively idle, thereby enhancing the efficiency of airport stand resources. On the other hand, it can increase the robustness of the stand allocation scheme. By allowing for idle time at stands, the system can better accommodate potential flight delays or early arrivals, thus reducing the sensitivity of the allocation plan to flight schedule changes and ensuring safety while mitigating delays.

$$\min Z_3 = \sum_{i \in F} \sum_{j \in F} \sum_{g \in G} I_{ij}^2 y_{ijg} \quad (3)$$

4.2.2. Constraints of the First-Stage Model

The constraints of the first-stage model add vehicle constraints on top of the basic constraints of the stand allocation.

$$\sum_{g \in G} x_{ig} = 1, \forall i \in F \quad (4)$$

$$\sum_{j \in F} y_{ijg} \leq x_{ig}, \forall i \in F, \forall g \in G \quad (5)$$

$$\sum_{i \in F} y_{ijg} \leq x_{jg}, \forall j \in F, \forall g \in G \quad (6)$$

$$S_i^F - M \times (1 - x_{ig}) \leq S_g^G, \forall i \in F, \forall g \in G \quad (7)$$

$$Sd_i x_{ig} + \delta^g - Sa_j x_{jg} + M \times (y_{ijg} - 1) \leq 0, \forall i, j \in F, \forall g \in G \quad (8)$$

$$Sd_i x_{ig} + \delta^g - Sa_j x_{jg} + M \times y_{ijg} \geq 0, \forall i, j \in F, \forall g \in G \quad (9)$$

$$|Sd_j - Sd_i| \geq \theta^t \times z_{ijt}, \forall i, j \in F, \forall t \in T \quad (10)$$

$$Sa_i + t_{rmg}^a + T^i + t_{grn}^d \leq Sd_i \quad (11)$$

Equation (4) guarantees that each flight has to be assigned to exactly one compatible stand. Equations (5) and (6) are introduced to ensure that exactly one flight is assigned immediately before a given flight j , and only one flight is assigned immediately after a given flight i . Equation (7) describes the size limitation between the stand and aircraft, which means that the size of the stand should be compatible with the size of the aircraft. Equations (8) and (9) guarantee that the interval between the arrival and departure of consecutive flights assigned to the same stand must be greater than or equal to the minimum safety interval for the stand. Equation (10) is the resource constraints; when different flights are assigned to the same stand serviced by a towing tractor, the planned departure time interval between two adjacent departing flights must be greater than the service interval of the towing tractor (including the minimum service time and the transfer time between stands). Equation (11) limits the time for the completion of ground support services for flights.

4.3. Multi-Type Ground Support Vehicle Scheduling Model

The sets, parameters and variables of the ground support vehicle scheduling model are shown in Table 3.

Table 3. Sets, parameters and variables of the ground support vehicle scheduling model.

	Notation	Meaning
sets	F^v	the set of flights requiring service from vehicle type v
	V	the set of vehicle types, $V \in \{1, 2, 3, 4, 5, 6\}$, represents ferry vehicles, passenger stairs, fuel trucks, water trucks, baggage belt loaders, and baggage tractors, respectively.
	T^v	the service time for vehicle type v .
parameters	s_{ij}^v	the driving distance for the vehicle from the location of flight i to the location of flight j .
	N_v	the time taken for the ferry vehicles to travel back and forth between the parking position g and the terminal.
	c	the number of vehicle type v .
	M	positive infinity.
	$[RT_i^v, DT_i^v]$	the time window during which vehicle type v can service flight i .
	C^4	the capacity of the water truck.
	d_i	the water demand for flight i .

Table 3. Cont.

	Notation	Meaning
variables	φ_{ijn}^v	binary decision variable, equal to 1 if the n -th vehicle of type v sequentially serves flights i and j , otherwise equal to 0.
	ϕ_n^v	binary decision variable, equal to 1 if the n -th vehicle of type v is in use, otherwise equal to 0.
	ψ_{in}^v	binary decision variable, equal to 1 if the n -th vehicle of type v serves flights i , otherwise equal to 0.
	w_{in}	binary decision variable, equal to 1 if the n -th water truck replenishes water resources after servicing flights i , otherwise equal to 0.
	ST_{in}^v	the time when the n -th vehicle of type v begins servicing flight i .
	C_{in}	the remaining quantity after the n -th water truck has finished servicing flight i .
	AT_{in}^v	the time when the n -th vehicle of type v arrives at flight i .

4.3.1. Vehicle Service Process and Synergy Analysis

- Passenger stairs: passenger stairs provide boarding and deboarding services for aircraft at remote stands, including for passengers, crew members, and cleaning staff. According to the regulations, the passenger stairs should arrive at the stand 5 min before the scheduled or estimated arrival time of the flight. The service time for the passenger stairs starts when the aircraft arrives at the stand and ends when boarding is completed. Since the service duration is relatively long and the stairs are large and difficult to maneuver, they are generally assigned to a specific apron for operations.
- Ferry vehicles: the ferry vehicles are used to transport passengers between remote stands and the terminal. For arriving flights, the first ferry vehicle should arrive at the assigned stand or designated waiting area at least 5 min before the scheduled/estimated arrival time of the flight, while subsequent ferry vehicles should be in position within 2 min after the departure of the first vehicle. For departing flights, depending on the aircraft type, the first ferry vehicle for C and D category aircraft should arrive at the boarding gate or designated waiting area no later than 45 min before the scheduled/target departure time; for E category aircraft, the first ferry vehicle should arrive no later than 50 min before the scheduled/target departure time, with other ferry vehicles arriving within 2 min after the departure of the previous vehicle. Different types of aircraft require a different number of ferry vehicles; for modeling purposes, the number of ferry vehicles required is set at 1, 2, and 3 for C, D, and E category aircraft, respectively.
- Fuel trucks: fuel trucks are categorized into tank trucks and pipeline trucks. Tank trucks are primarily used in medium and small airports that lack underground fuel pipelines. Pipeline trucks do not have the function of transporting and storing fuel; they are more convenient and safer and are primarily suitable for large airports with underground fuel pipelines. Fueling should occur after the jet bridge or passenger stairs have been connected and confirmed by the airline representative. The airline representative is required to provide the necessary fuel amount within 5 min of the fuel truck's arrival. Fueling is generally completed 5 min before boarding begins; this study does not consider passenger boarding during fueling or special circumstances.
- Water trucks: water trucks supply drinking water to aircraft. The operation can begin once the jet bridge or passenger stairs are connected, and it must be completed no later than 15 min before the scheduled/target departure time. Water trucks have limited capacity and require resource replenishment at supply points. Since the aircraft's turnaround time is generally longer than the service time, the service window for water trucks is more flexible compared to other ground support vehicles.
- Baggage belt loaders and baggage tractors: baggage tractors and belt loaders are responsible for the transportation and loading/unloading of baggage. Baggage tractors shuttle between the baggage handling center and the apron to transport baggage, while belt loaders use conveyor belts to access the elevated cargo hold, assisting staff with

loading and unloading. Together, baggage tractors and belt loaders complete the baggage handling tasks for the aircraft. Baggage and cargo handling staff, along with the equipment, should be in position 5 min before the scheduled/estimated arrival time, and the loading of cargo and baggage must be completed 5 min before the scheduled/target departure time.

Based on the task associations of vehicles, ground support vehicles can be divided into three types of collaborative relationships:

(1) Sequential Task Relationships

Sequential task relationships are characterized by a clear time order between tasks. For example, ferry vehicles and fuel trucks, as fueling is generally not permitted while passengers are on board. Therefore, after the ferry vehicle completes its task, the fuel truck can service the aircraft. The fueling task must be completed before the ferry vehicle for departing passengers arrives.

(2) Parallel Task Relationships

Parallel task relationships mean there are no constraints on the order of task execution. For example, water trucks, garbage trucks, and catering trucks operate independently and do not need to follow one another in a specific order.

(3) Joint Task Relationships

Joint task relationships refer to situations where multiple vehicles must work together to complete a single task, with each being indispensable. For example, baggage tractors and baggage belt loaders work in tandem to handle baggage loading and unloading. Since the baggage tractor needs to shuttle between the baggage handling center and the apron, which takes some time, the baggage belt loader should arrive at the designated task point before the baggage tractor to avoid delays. Once the baggage handling is completed, the tasks for both types of vehicles are finished.

4.3.2. Objectives of the Second-Stage Model

1. Minimize the driving distance of ground support vehicles.

The objective function is shown in Equation (12). Minimizing the driving distance aims to reduce vehicle operating costs and avoid extensive movement of vehicles within the operational area, thereby enhancing safety. Since vehicles such as water trucks and baggage tractors need to travel to resource replenishment points or baggage handling centers, the driving distance includes the distances traveled by vehicles to other service locations between their assignments.

$$\min Z_1 = \sum_{v \in V} \sum_{n \in N_v} \sum_{i,j \in F} s_{ij}^v \cdot \varphi_{ijn}^v \quad (12)$$

2. Minimize the number of ground support vehicles used.

The objective function is shown in Equation (13). Minimizing the number of vehicles used not only aims to enhance service efficiency and explore the minimum resource requirements for completing support tasks but also serves to reserve vehicles within limited resources to respond to unforeseen circumstances.

$$\min Z_2 = \sum_{v \in V} \sum_{n \in N_v} \phi_n^v \quad (13)$$

3. Minimize the waiting time of ground support vehicles.

The objective function is shown in Equation (14). Minimizing waiting times helps prevent vehicles from staying in the same location for extended periods, thereby increasing their effective usage time.

$$\min Z_3 = \sum_{v \in V} \sum_{n \in N_v} \sum_{i \in F} (ST_{in}^v - AT_{in}^v) \quad (14)$$

4.3.3. Constraints of the Second-Stage Model

The constraints of the second-stage model include both the common vehicle constraints and the coordination constraints between different vehicle types.

$$\sum_{n \in N_v} \psi_{in}^v = 1, \forall i, j \in F^v, \forall v \in V \quad (15)$$

$$\sum_{i \in F^v} \varphi_{ijn}^v - \sum_{k \in F^v} \varphi_{ikn}^v = 1, \forall j \in F^v, \forall v \in V, \forall n \in N_v \quad (16)$$

$$\sum_{n \in N_v} \phi_n^v \leq N_v, \forall v \in V \quad (17)$$

$$ST_{in}^v + T^v + \frac{s_{ij}^v}{c} - M(1 - \varphi_{ijn}^v) \leq ST_{jn}^v, \forall i, j \in F^v, \forall n \in N_v, \forall v \in V \quad (18)$$

$$RT_i^v \cdot \psi_{in}^v \leq ST_{in}^v \cdot \psi_{in}^v \leq DT_i^v \cdot \psi_{in}^v, \forall i \in F^v, \forall n \in N_v, \forall v \in V \quad (19)$$

$$AT_{in}^1 \geq AT_{in}^2, \forall i \in F^v, \forall n \in N_v \quad (20)$$

$$ST_{in}^3 \geq ST_{in}^5 + T^5, \forall i \in F^a, \forall n \in N_v \quad (21)$$

$$ST_{in}^5 \geq ST_{in}^3 + T^3, \forall i \in F^d, \forall n \in N_v \quad (22)$$

$$0 \leq C_{in} \leq C^4 \quad (23)$$

$$C_{jn} \geq C_{in} - M(1 - \varphi_{ijn}^v + \phi_n^v) - d_j, \forall i, j \in F^v, \forall n \in N_v \quad (24)$$

$$C_{jn} \geq C^4 - M(1 - \varphi_{ijn}^v + \phi_n^v) - d_j, \forall i, j \in F^v, \forall n \in N_v \quad (25)$$

$$AT_{in}^6 \geq AT_{in}^5, \forall i \in F^v, \forall n \in N_v \quad (26)$$

$$ST_{in}^5 = ST_{in}^6 = \begin{cases} AT_{in}^6 & AT_{in}^6 \geq RT_i^v \\ RT_i^v & AT_{in}^6 < RT_i^v \end{cases} \quad (27)$$

Equation (15) guarantees that each flight must be served by one vehicle for each type of service. Equation (16) guarantees that each flight has a predecessor and a successor in the sequence served by one vehicle. Equation (17) limits the available number of vehicles. Equation (18) is the serving sequence constraint on two adjacent flights served by one vehicle. Equation (19) means that the time a vehicle begins servicing a flight must be within the flight's service time window. Equation (20) is the collaborative constraint between ferry vehicles and passenger stairs, indicating that the passenger stairs must arrive before the ferry vehicles. Equations (21) and (22) are the collaborative constraints between the ferry vehicle and the fuel truck. For arriving flights, the fuel truck must begin its service after the ferry vehicle has completed its task. For departing flights, the fuel truck must finish its service before the ferry vehicle starts its operation. Equation (23) constrains the water is restricted to be no more than C^4 . Equation (24) constrains that if the water truck does not return to the resource supply point to refill after serving the flights i , the water remaining in that vehicle needs to meet the restriction. Equation (25) constrains that if the water truck returns to the resource supply point to refill after serving the flights i , the water fluid remaining in that vehicle needs to meet the restriction. Equations (26) and (27) are the collaborative constraints between the baggage belt loader and the baggage tractor,

indicating that the baggage belt loader must arrive before the baggage tractor, and service starts after both the baggage belt loader and the baggage tractor have arrived.

5. The NSGA-II Algorithm Combining Local Search Strategies

Genetic algorithm (GA) is a computational optimization technique inspired by the principles of biological evolution and natural selection. It mimics the genetic processes found in nature, such as reproduction, mutation, and crossover, to iteratively improve solutions to complex problems. Based on the traditional NSGA-II algorithm, we incorporate destruction and repair operations with a local search. After completing the crossover and mutation operations of the genetic algorithm, destruction and repair operations are used to adjust individuals within the population, ensuring that the algorithm can better explore optimal solutions while maintaining strong global search capabilities. Taking the stand allocation problem as an example, the modeling steps are explained as follows.

(1) Chromosome encoding. An integer encoding method is adopted. The genes in the chromosome are sorted by the scheduled arrival time of flights. The numbers on the genes represent the stand number. Figure 4 shows a part of the chromosome.

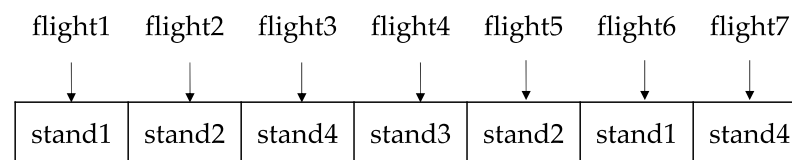


Figure 4. Chromosome encoding for stand allocation problem (example).

(2) Generate the initial population. An initial population is generated using a random constraint-satisfying method. The specific steps are as follows:

Step 1: Select the first flight $i, i \in F$ from the set of unallocated flights F . According to the aircraft type of flight i , we can obtain the set of compatible stands G_i ;

Step 2: If there is a free stand g , assign flight i to stand g . Otherwise, traverse these stands and check if the time between the completion of the previous flight's service at the stand and the arrival time of the current flight meets the required time interval constraint. Add the stands that meet the constraint to the feasible stand set. Randomly select the smallest stand from the feasible set and assign the flight to this stand. If the feasible stand set is empty, traverse the set G_i and choose the stand that completes its task the earliest.

(3) Evolve the population using the two-point crossover method and random mutation method to obtain offspring individuals. In the initial stage of the population, the crossover probability and mutation probability are set relatively high, while in the later stages of iteration, they should gradually decrease. The specific probability definition is shown in Equation (28).

$$p_i = p_{max} - i \times \frac{p_{max} - p_{min}}{N} \quad (28)$$

where i is the current iteration generation, p_i is the crossover/mutation probability for the current iteration, p_{max} and p_{min} are the initially set maximum and minimum values, respectively, and N is the population size.

(4) Local search: First, apply the destroy method (function $d(x)$) to the current chromosome to randomly select flights for removal. Then, use the repair method (function $r(x)$) to creating a new solution.

Destroy Method: select m flights to remove from the current solution X_0 and add them to the removed flight set A . The first flight in A will be the flight with the maximum delay from the current solution X_0 . The selection method for the remaining $m - 1$ flights is as follows: randomly select a flight from A , then sort the remaining flights in X_0 based on

their relevance to this selected flight to obtain the set X' [36]. The relevance calculation method is as Equations (29)–(30):

$$R_{ij} = \frac{1}{d'_{ij} + v_{ij}} \quad (29)$$

$$d'_{ij} = \frac{d_{ij}}{\max D} \quad (30)$$

$$v_{ij} = \begin{cases} 1, & \text{Flights } i \text{ and } j \text{ are assigned to the same stand} \\ 0, & \text{otherwise} \end{cases} \quad (31)$$

Equation (30) explains the method of standardizing d_{ij} , where d_{ij} is the distance between flights i and j , $\max D$ represents the largest distance. The closer two flights are, the higher their correlation. To prevent the algorithm from overly relying on this correlation, a random operator is introduced to enhance the algorithm's diversity in Equation (32). By incorporating randomness, the algorithm avoids becoming too deterministic in its decisions, allowing for a broader exploration of the solution space and potentially avoiding local optima. This balance between correlation-based decisions and randomness helps the algorithm maintain flexibility and adaptability throughout the optimization process.

$$\lambda = \text{random}([0, 1])^D \times n' \quad (32)$$

where n' represents the number of remaining flights in the set X' . The λ -th flight from the set X' is chosen to be removed and added to set A . It can be observed that as $D \rightarrow \infty$, the selected flights have a stronger correlation.

Repair method: flights from set A are randomly assigned to compatible stands. The fitness value of all feasible solutions is calculated, and the solution with the best fitness value is selected as the new feasible solution. This ensures that after the destroy phase, the repaired solution is optimized for fitness, maintaining the integrity of the overall search process.

(6) Fitness value evaluation: To assess the feasibility of the solution after the local search, the fitness value is used.

$$\text{fitness}(x) = Z1 + Z2 + Z3 + \omega W \quad (33)$$

Equation (33) calculate the fitness, $Z1$, $Z2$, $Z3$ represent the values of the objective functions. W is the total delay time of all flights, and ω is the penalty value, empirically set to 100. The fitness function, therefore, incorporates both the objective function values and the total delay, with the penalty value ensuring that excessive delays are strongly discouraged.

(7) Merge offspring and parent populations and select individuals based on fast non-dominated sorting and crowding distance calculation. Identify the non-dominated solution set P_m from the population, assign the current rank as $\text{rank}(i) = 1$, and add these solutions to the non-dominated layer F . Repeat this process iteratively, assigning a non-dominated rank value to all individuals until the entire population is sorted. For individuals with the same $\text{rank}(i)$, the crowding distance is calculated based on Equations (34) and (35).

$$d_i = \frac{d_i + (v_{i+1}^m - v_{i-1}^m)}{v_{\max}^m - v_{\min}^m} \quad (34)$$

$$d_0 = d_n = M \quad (35)$$

where $d_i (i \in [1, n - 1])$ represents the crowding distance of an individual, with an initial value of 0, v_{i+1}^m denotes the function value of the m -th objective for the individual, $v_{\max}^m$ and $v_{\min}^m$ are the maximum and minimum values of the m -th objective, respectively, M is a very large number.

(8) Termination condition: To improve the computational efficiency of the algorithm, the number of iterations is set as the termination condition.

LS-NSGA-II algorithm pseudocode see in Algorithm 1.

Algorithm 1. Pseudocode of LS-NSGA-II

Input Initial population P_0 , population size N , maximum iterations G_{max} , crossover probability P_c , mutation probability P_m , flight set F , stand set G

Output The population P after iteration

```

1 Perform fast non-dominated sorting and crowding distance assignment for  $P$ 
2  $g \leftarrow 1$ 
3  $P \leftarrow P_0$ 
4 while  $g < G_{max}$ 
5    $F \leftarrow \emptyset$  //Generate the parent population  $F$ 
6   For  $i \leftarrow 1$  to  $length[P_0]$ 
7     Select individual  $x$  from  $P$  using binary tournament selection
8     Put  $x$  into  $F$ 
9   end for
10  Perform crossover and mutation operations on  $F$  to generate the offspring population  $C$ .
11  For  $i = 1$  to  $length[C]$ 
12     $x = C(i)$ 
13     $x' \leftarrow r(d(x))$  //Perform repair and destruction operations on
                           individual  $x$  to generate new individual  $x'$ .
14     $C(i) = x'$ 
15  end for
16   $R \leftarrow C \cup F$  //Merge parent and offspring populations
17  Perform fast non-dominated sorting and crowding distance assignment for  $R$ 
18  Select  $N$  individuals from  $R$  to form the new population  $P$ 
19   $g \leftarrow g + 1$ 
20 end while
21 return  $P$ 

```

6. Numerical Study

6.1. Experimental Data and Settings

In order to verify the proposed two-stage optimization model, one of the busiest hub airports in the world, Shenzhen Bao'an International Airport, is used as the research object. We consider the actual operational data of the flight, which involves a total of 82 pairs of flights in the time range of 17:00–20:00. All flights are also divided into three categories based on the aircraft wingspan length, including 59 flights for category C, 17 flights for category D, and 6 large flights for category E. The time is converted into minutes based on the difference from 17:00. More detailed flight information is shown in Table 4.

Table 4. Basic information of flight data.

Flight	Aircraft Type	Arrival Runway	Departure Runway	Arrival Time	Departure Time	Arrival Runway
0	3	15	15	0	125	15
1	1	16	15	0	80	16
2	1	16	15	0	80	16
3	1	16	15	0	85	16
...	...	...	...	...	...	...
79	1	15	16	185	270	15
80	2	16	15	190	270	16
81	2	16	15	195	285	16

A total of 31 stands at the airport were selected for the stand allocation model. According to the size of aircraft wingspan that the stand can accommodate, all stands are classified into three categories: 11 stands for category E, 1 stand for category D, and 19 stands for

category C. Among these, 22 are contact stands, and 9 are remote stands. The taxi-in and taxi-out times for different aircraft types are derived from historical data. All stands are self-maneuvering in and pushback-out, requiring tug services for pushback operations. Figure 5 shows the taxi time of Class C and Class E aircraft from different runways and different stands.

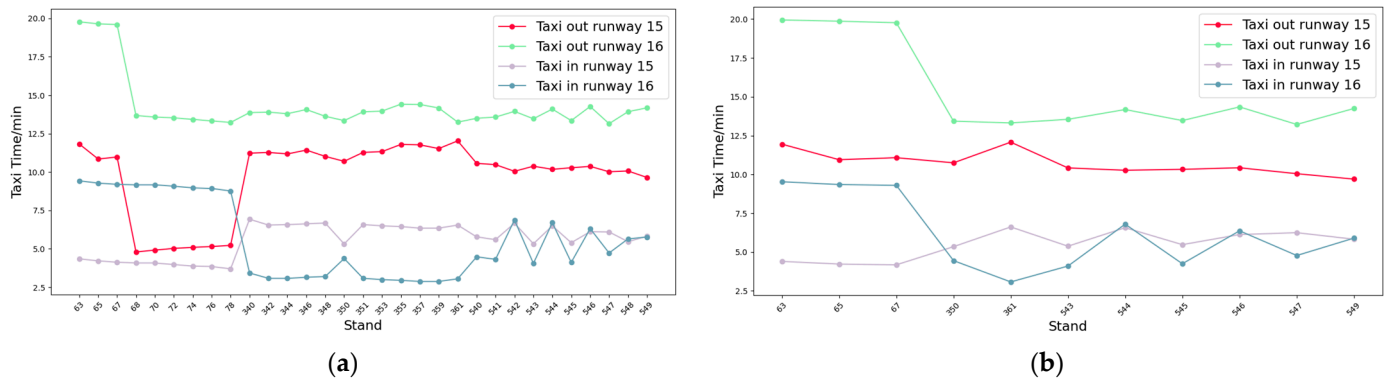


Figure 5. Taxi times for different types of aircraft. (a) C-type aircraft; (b) E-type aircraft.

Based on the field investigation at Shenzhen Airport, Table 5 presents the specific parameter settings for vehicles used in the experiments of this study.

Table 5. Vehicle information settings table.

Vehicle Information	Value
Number of ferry vehicles	10
Number of passenger stairs	10
Number of fuel trucks	12
Number of water trucks	8
Number of baggage tractors and belt loaders	60
Passenger boarding/un boarding time	5 min
Fuel truck service time	C: 17 min D: 22 min E: 25 min
Water truck service time	C: 5 min D: 7 min E: 10 min
Baggage tractor and belt loaders service time	Arrival: 15 min Departure: 25 min
Vehicle driving speed	30 km/h

The LS-NSGA-II algorithm parameters are shown in Table 6. All the models related to this paper were created on a PC consisting of an Intel Core i7-10700H@2.90 GHz CPU and 16 GB RAM. With PyCharm as the implementation platform, Python 3.12.3 was used as the core tools to solve the Two-Stage Optimization Model and obtain the optimal results of stand allocation and ground support vehicle scheduling.

Table 6. The LS-NSGA-II algorithm parameters.

Maximum Iterations	Population Size	Crossover Probability	Mutation Probability	Objective Function Weights
800	100	$P_{C_{max}}$: 0.8 $P_{C_{min}}$: 0.7	$P_{m_{max}}$: 0.3 $P_{m_{min}}$: 0.2	1/3

6.2. Results and Discussion

6.2.1. Stand Allocation Problem

Figure 6 shows the Pareto front of the stand allocation problem solved by the LS-NSGA-II model. Considering three objectives: taxi fuel consumption, ferry vehicle driving time, and the sum of the squares of the idle times, the model provides the optimal solutions for different objective functions after iterations. In Figure 6, Z_1 represents the taxi fuel consumption, Z_2 represents the ferry vehicle driving time, and Z_3 represents the sum of the squares of the idle times.

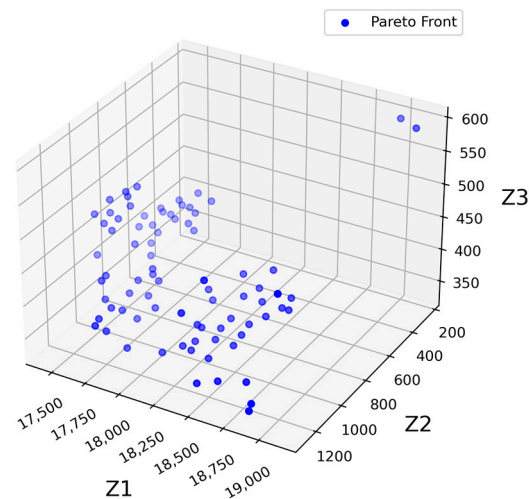


Figure 6. Pareto front of the stand allocation problem.

Table 7 provides the results of some solutions from the Pareto front. Solution 1 will serve as the input for ground support vehicle scheduling. Figure 7 shows the convergence of fitness values under NSGA-II and LS-NSGA-II. We can find that in the early stages of algorithm iteration, the LS-NSGA-II algorithm is able to find the optimal solution more quickly; in the later stages, it escapes local optima and converges earlier.

Table 7. The results of some solutions from the Pareto front.

	Taxi Fuel Consumption	Ferry Vehicle Driving Time	Idle Time
1	17,969.15	760	452.77
2	18,989.89	240	595.2
3	19,131.96	280	594.73
4	17,940.98	720	471.13

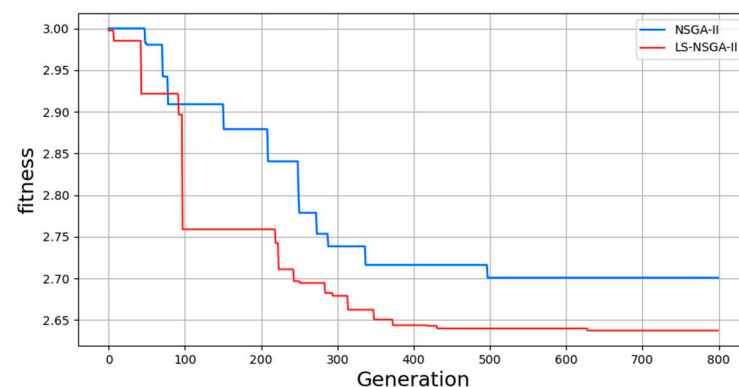


Figure 7. Fitness value iteration for stand allocation problem under NSGA-II and LS-NSGA-II.

The Gantt chart in Figure 8 illustrates the final stand allocation scheme. The taxi fuel consumption is 17,969.15 kg, the ferry vehicle driving time is 760, and the squared stand idle time is 452.77. Each segment in the chart represents the time a stand is occupied by a flight. It can be seen from the chart that flights between 50 and 220 min are dense, and the allocation of the contact stands is relatively balanced in this scheme.

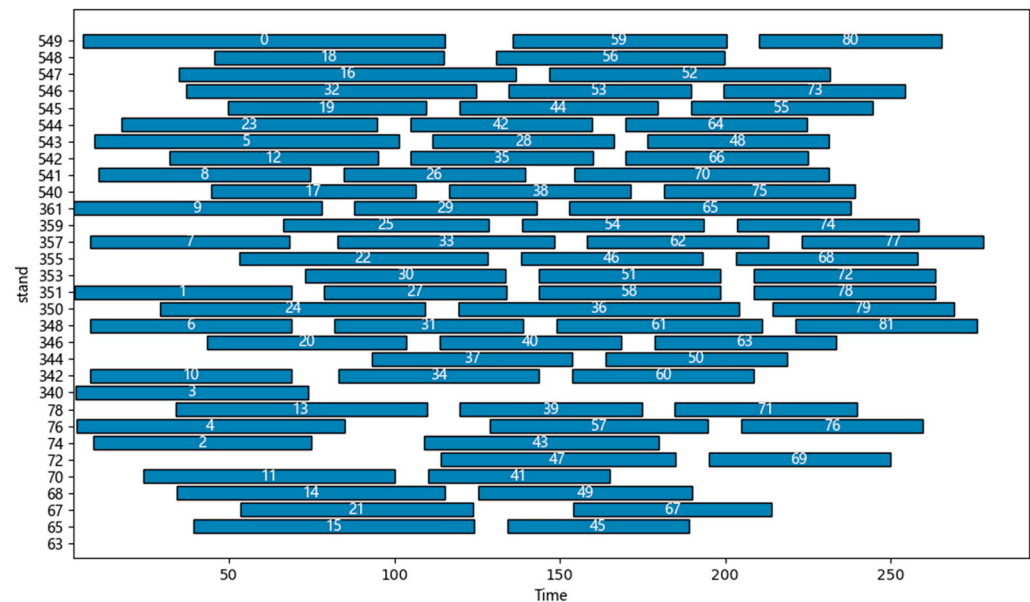


Figure 8. Gantt chart of stands.

6.2.2. Ground Support Vehicle Scheduling

Table 8 shows a comparison of scheduling results between the FCFS strategy and LS-NSGA-II algorithms. From the table, it is evident that the LS-NSGA-II algorithm significantly improves upon the FCFS approach, particularly in optimizing waiting time and driving distance. The model improved vehicle utilization efficiency by 37.5%, reduced travel distance by 20.4%, and decreased waiting time by 57.6%. Due to the demand of flights and the number of tasks, the reduction in the number of some types of vehicles, like ferry vehicles and passenger stairs, is not significant. Notably, the number of baggage belt loaders and baggage tractors shows a significant optimization, as only about half the number of vehicles scheduled under the FCFS approach is required to meet the demand.

Table 8. Comparison of scheduling results between the FCFS strategy and LS-NSGA-II algorithms.

Vehicle Type	Vehicle Quantity		Driving Distance/m		Waiting Time/min	
	FCFS	LS-NSGA-II	FCFS	LS-NSGA-II	FCFS	LS-NSGA-II
Ferry vehicle	10	9	397,253.81	341,714.73	293.99186	79.44
Passenger stair	10	8	26,395.68	20,953.97	285.08	157.77
Fuel truck	12	8	138,452.78	87,697.23	852.76	100.32
Water truck	8	4	142,632.58	95,009.27	896.53	63.02
Baggage belt loader	60	35	269,744.42	196,246.44	4372.17	2359.20
Baggage tractor	60	36	929,055.84	772,663.73	6545.82	2852.21

Figures 9 and 10 compare the results of FCFS and LS-NSGA-II using examples of water trucks. In Figure 9a, since LS-NSGA-II uses fewer vehicles than FCFS, each vehicle handles more flight tasks than in FCFS, resulting in a greater distance of some water trucks. However, the total distance covered by all vehicles is less than that of FCFS, indicating a more efficient task allocation for vehicles and reducing frequent terminal trips. In Figure 9b, the reasonable allocation significantly reduces the waiting time of vehicles, with some

vehicles even achieving a waiting time of zero. Figure 10 illustrates the distribution of vehicle tasks on the apron in the FCFS and LS-NSGA-II scheduling schemes. Apron 1 and 2 are located in the center of the airport, separated from Apron 3 and 4 by one runway. Therefore, vehicles move shorter distances between Apron 1 and 2 or 3 and 4, while moving around the runway to service different aprons results in longer distances. In the FCFS scheme, most vehicles need to cross three to four aprons and frequently shuttle between both sides of the runway to enter different aprons. In contrast, in the LS-NSGA-II scheme, water trucks primarily serve the same apron continuously on one side of the runway, resulting in less distance traveled.

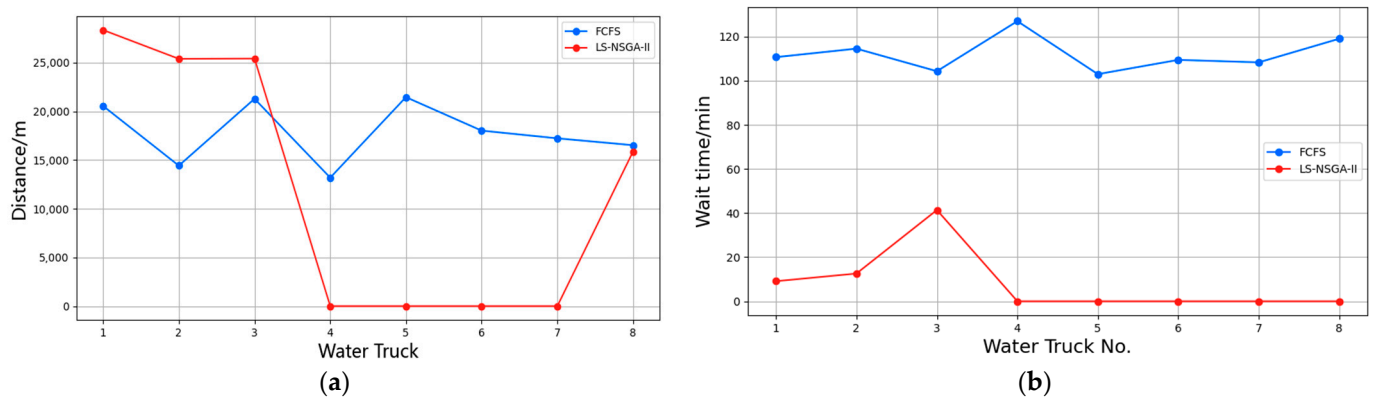


Figure 9. Distribution of water truck distance and wait time using FCFS and LS-NSGA-II. (a) Distance of water trucks; (b) Wait time of water trucks.

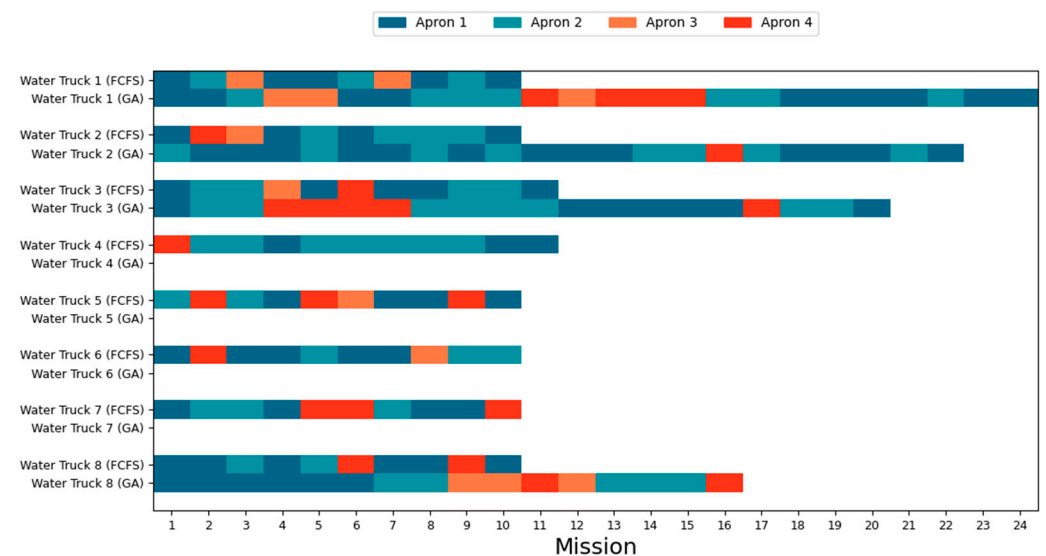


Figure 10. Distribution of Vehicle Tasks on the Apron using FCFS and LS-NSGA-II.

Taking the ferry vehicles and fuel truck as examples, Figure 11 illustrates the vehicle scheduling results. In Figure 11a, “x-arr” indicates that the ferry vehicle arrives at the stand of flight x to take arriving passengers to the exit, while “dep-x” indicates that the ferry vehicle takes departing passengers from the boarding gate to the stand of flight x.

From the vehicle Gantt chart and the comparison between the algorithm’s optimal solution and the FCFS results, the characteristics of different vehicles executing tasks can be observed. The ferry vehicles and passenger stairs are involved in passenger services and are closely linked to the aircraft’s pushback times. Consequently, the time windows for these two types of vehicles are relatively tight, and their operational flexibility is limited. Therefore, the optimization of travel distance and waiting time for the dispatch of ferry

vehicles and passenger stairs is more pronounced. Constrained by the nature of the tasks, the optimization of vehicle numbers compared to the FCFS approach is relatively minor.

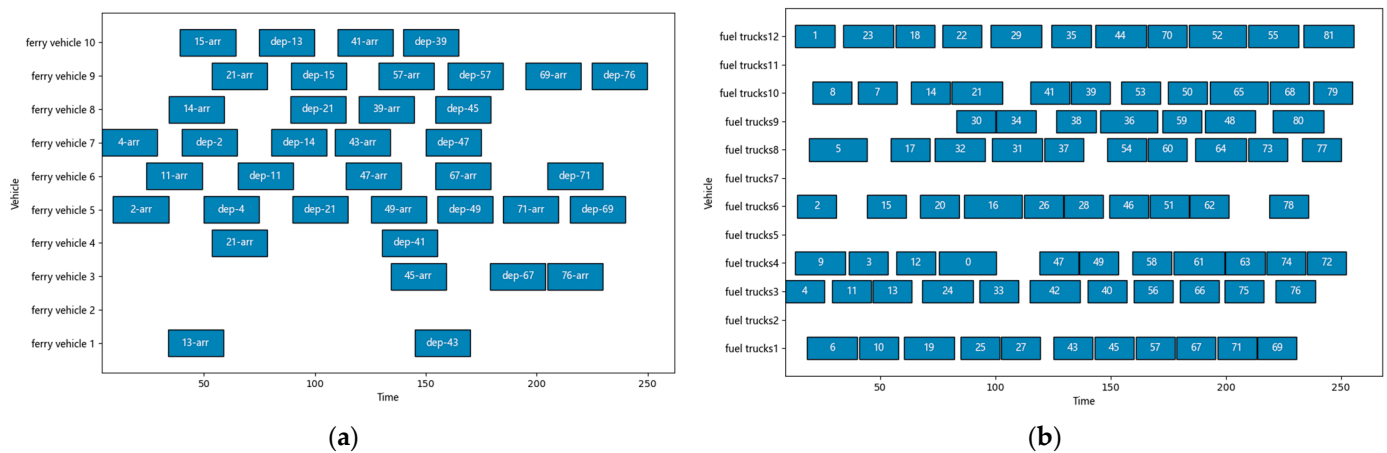


Figure 11. Gantt chart of ground support vehicles. (a) Ferry vehicles; (b) Fuel trucks.

Fuel trucks are influenced by ferry vehicle tasks. In this study, all fuel truck tasks are non-passenger fueling. Thus, the time windows for fuel trucks occur after the ferry vehicles servicing arriving flights have completed their tasks and before the ferry vehicles servicing departing flights begin. This provides a relatively ample timeframe. The Gantt chart for fuel trucks shows that they prefer to execute tasks consecutively at adjacent stands or within the same apron, significantly reducing both driving distances and waiting times.

Water trucks, due to capacity limitations, must return to the resource point to replenish water after serving a certain number of flights. Their service time is short, and their time window is broader, allowing for the greatest operational flexibility. As a result, water trucks are the least numerous among all vehicles. Under conditions where resource capacity is met, water trucks, like fuel trucks, also prefer to perform tasks consecutively at adjacent stands or within the same apron.

For baggage handling tasks, the related vehicles (baggage belt loaders and baggage tractor) must carry out both loading and unloading operations. Baggage tractors need to travel back and forth between the baggage handling center and the stand, resulting in longer driving distances. Baggage conveyor belts have to wait for the baggage tractors to perform their tasks together. Due to the lengthy loading and unloading times, there is significant demand for baggage belt loaders and baggage tractors.

6.3. Sensitivity Analysis

To understand the influence of different minimum turnaround times, four additional scenarios are introduced, namely, -10 , -5 , $+5$, and $+10$, representing changes from the original minimum turnaround time of C-type aircraft. The time of aircraft in each scenario is shown in Table 9.

Table 9. Testing instances for different scenarios of C-type aircraft.

Instances	-10	-15	Base Case	$+5$	$+10$
C-type aircraft	35	40	45	50	55
D-type aircraft	60	60	60	60	60
E-type aircraft	75	75	75	75	75

Figure 12 illustrates the results across different scenarios. As the turnaround time for C-type aircraft increases from 35 to 55 min, taxi fuel consumption initially decreases before rising again. This trend occurs because some flights using Runway 15 are allocated stands slightly farther from the runway than remote stands. Consequently, at a turnaround time of

30 min, fuel consumption experiences a slight increase. The driving time of ferry vehicles increases with longer turnaround times, as more flights are allocated to remote stands. After the turnaround time reaches 45 min, more D-type and E-type aircraft, which require additional ferry vehicles, are allocated to remote stands. Idle time shows minimal variation across scenarios and trends opposite to the other two objectives. Shorter turnaround times lead to longer intervals between flights, while longer turnaround times result in only necessary safety intervals. These findings suggest that changes in turnaround time mainly impact the allocation of aircraft to remote stands, thereby influencing taxi fuel consumption and parking stand idle times.

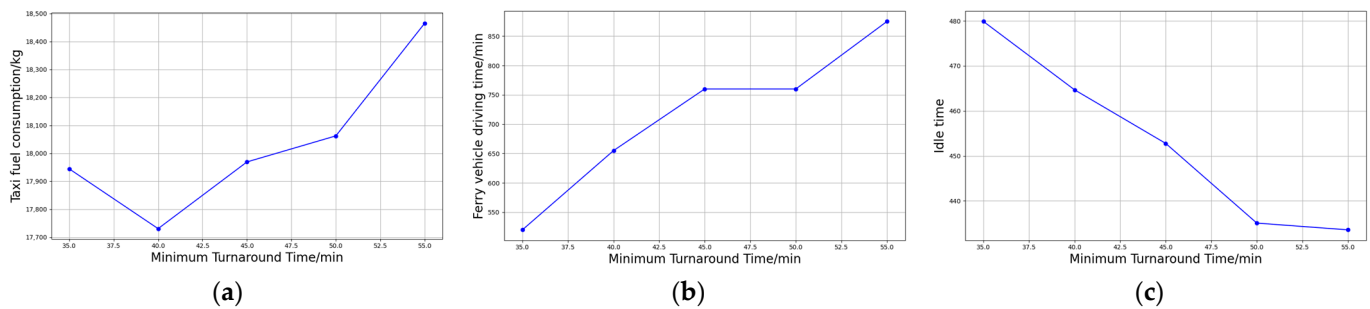


Figure 12. Test results for different minimum turnaround time of C-type aircraft. (a) Test results of taxi fuel consumption; (b) Test results of driving time of ferry vehicles; (c) Test results of idle times.

7. Conclusions

This study proposes a two-stage optimization model for scheduling airport operational and support resources. In addressing the stand allocation problem, we have incorporated considerations for the use of tractors and ferry vehicles, optimizing stand allocation results and pushback times, which serve as input for the second stage of support resource scheduling optimization. For the scheduling of support vehicles, we optimized task assignments and driving routes based on the collaborative relationships among vehicles. By addressing the stand allocation problem with regard to transient resources, we can mitigate delays caused by service conflicts among vehicles, while the optimized scheduling of support vehicles based on stand allocation results can enhance the utilization efficiency of existing airport support resources. The model we proposed is based on flight schedules to generate reasonable gate allocation plans and vehicle scheduling solutions, achieving airport resource scheduling and allocation during the pre-tactical phase. The research findings offer meaningful insights for the transient support of flights, improving the efficiency, safety, and effectiveness of airport ground operations, and effectively integrating existing resources. It can be regarded as a powerful decision-support tool for air traffic controllers and airport ground staff.

Our proposed two-stage model assumes fixed driving routes and does not account for potential route conflicts among aircraft, vehicles, and between aircraft and vehicles. In future work, we will focus on further investigating the taxiing trajectories at the airport and the integrated scheduling and management of airport ground operations. Additionally, in scheduling vehicles, we did not consider personnel allocation. Future research could explore “human–aircraft–vehicle” multi-agent collaborative scheduling to promote the development of smart airports. To more comprehensively account for the actual operation of airports, future research could include different operational scenarios caused by weather conditions and emergencies, enabling a deeper exploration of the algorithm’s performance under various scenarios. With the continuous development of artificial intelligence and machine learning technologies, future research can leverage these technologies to simulate and predict airport operation scenarios. These approaches can make the data more aligned with the actual conditions of airport operations, thereby improving the optimization of resource allocation methods.

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